The EcoliGenes library

Solving the never-ending struggle with *Escherichia coli* K-12's genes

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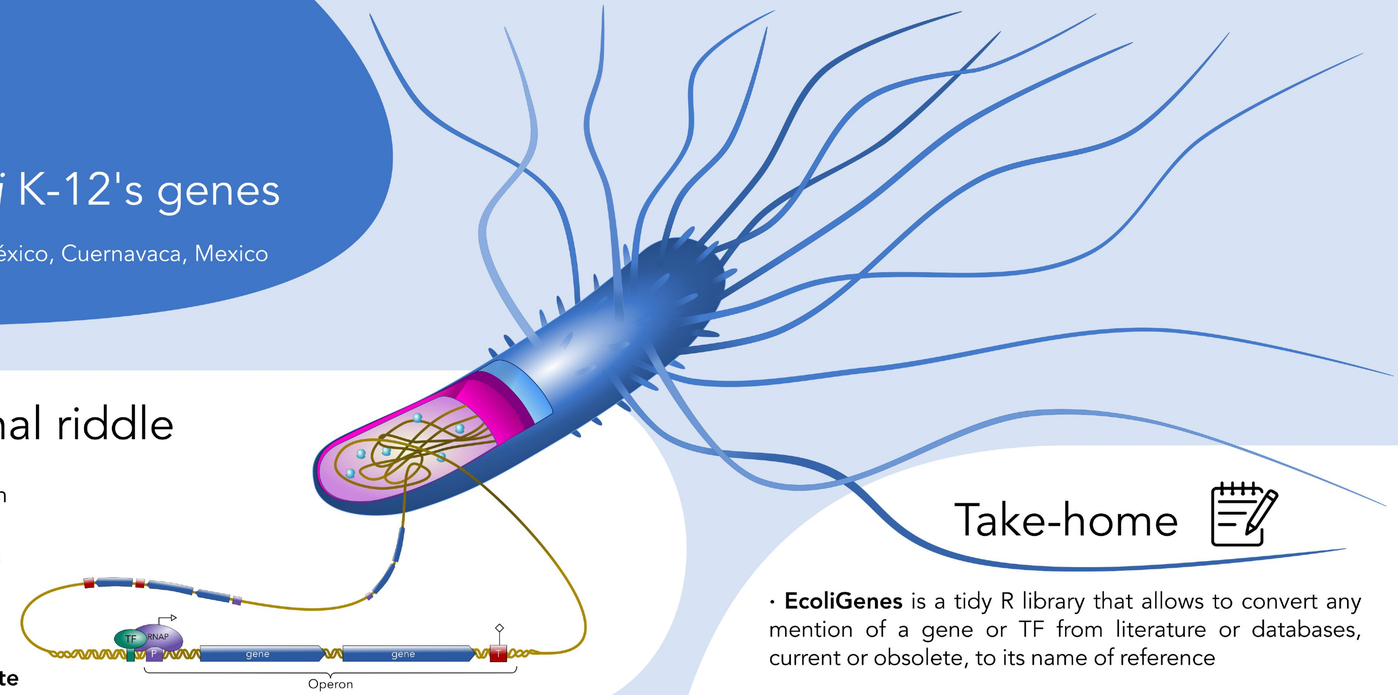
Escherichia coli K-12: a well known model, yet an eternal riddle

- *E. coli* is a classic **model organism** that has been used for decades in genetics and transcriptomics research
- Genome of 4.6 Mb fully sequenced and published in 1997
- Features more than **4,000 genes** organized in operons, defined by transcription start and termination sites
- An estimated **300 trancription factors** (TFs) control its densely inter-connected regulatory network

- Hundreds of genes are still not characterized, and most TFs have unknown function and targets
- Annotations keep adding up over time, leading to a **large amount of information** that is **tough to navigate**

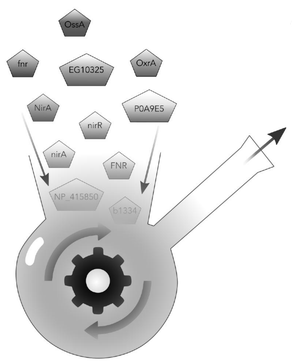


Take-home

- **EcoliGenes** is a tidy R library that allows to convert any mention of a gene or TF from literature or databases, current or obsolete, to its name of reference

- **Genes, coordinates** and other **genomic features** can be manipulated, converted, and summarized just like the next dataframe, using packages from the tidyverse

A neat & exhaustive collection of names and synonyms



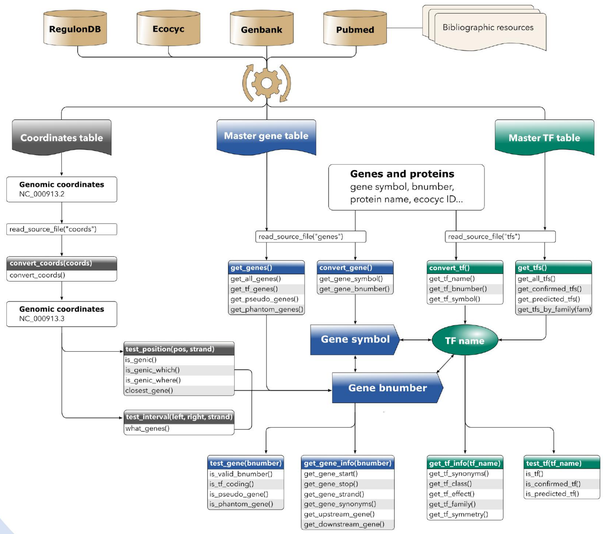
fnr | b1334 | b1334 | fnr | ECK120000319 | EG10325 | ossA | oxrA | nirA | nirR | 1,398,774 | 1,399,526 | reverse | FNR | FNR | Fnr | OssA | OxrA | NirA | NirR | PF00027 | PF00325 | NP_415850 | P0A9E5 | DNA-binding transcriptional dual regulator FNR | Global Regulator | CRP family | covalent

lrp	b0889	alsB, b0889, ECK120000540, EG10547, ihb, livR, lrp, lrs, lss, lstR, mbf, oppI, rblA	932,595	933,089	forward
dgoR	b4479	b3694, b3695, b4479, dgoR, ECK120004308, G7789, yidW	3,874,471	3,875,160	reverse
gyrA	b2231	b2231, ECK120000416, ECK2223, EG10423, gyrA, hisW, nalA, nfxA, norA, parD	2,866,559	2,867,551	reverse
rpoS	b2741	abrD, appR, b2741, csi2, dpeB, ECK120000503, ECK2736, EG10510, katF, nur, otsX, rpoS, sigS	2,336,793	2,339,420	reverse

Lrp	AlsB, ECK120005016, Ihb,LivR, Lrp, Lrs, Lss, LstR, Mbf, NP_415409, Oppl, P0ACJ0, RblA	DNA-binding transcriptional dual regulator Lrp Global Regulator AsnC family allosteric
DgoR	DgoR, ECK120008747, G7789, P31460, YidW, YP_026239	DNA-binding transcriptional regulator DgoR Local Regulator

- After querying several databases, **gene and TF names, synonyms and IDs** are gathered and organized
- Genes are indexed using a **reference symbol** and bnumber, TFs are indexed by their common name
- Alternative names and synonyms are kept in a **dictionary**, along with additional attributes

The complete framework



Use cases

- A random list of gene names is generated
- Genes are translated to their reference names and filtered
- Some of their attributes are retrieved

```
gene_list <- sample(get_all_genes_synonyms(), 100)          # get a random list of gene names

gene_table <- data.frame(gene_list) %>%
  mutate(symbol = convert_gene(gene_list, to = "symbol"),   # get symbol of reference
         bnumber = convert_gene(gene_list, to = "bnumber"), # get bnumber of reference
         start = get_gene_start(bnumber),                   #
         stop = get_gene_stop(bnumber),                     # get gene coordinates & strand
         strand = get_gene_strand(bnumber)) %>%             #
  filter(is_tf_gene(bnumber)) %>%                           # select only TF-coding genes
  mutate(tf_name = convert_tf(bnumber),                     # get TF common name
         family = get_tf_family(tf_name))                   # get TF attributes
```

gene_list
ydhM
G7326
yhiT
yeaM
ECK120000526
rtcR
ECK4055
b2087

gene_list	symbol	bnumber	start	stop	strand	tf_name	family
ydhM	nemR	b1649	1,726,023	1,726,622	+	NemR	TetR/AcrR
G7326	iscR	b2531	2,661,643	2,662,131	-	IscR	Rrf2
yhiT	gadE	b3512	3,658,366	3,658,893	+	GadE	LuxR/UhpA
yeaM	nimR	b1790	1,874,755	1,875,576	-	NimR	AraC/XylS
ECK120000526	lexA	b4043	4,257,115	4,257,723	+	LexA	LexA
rtcR	rtcR	b3422	3,558,268	3,559,866	+	RtcR	EBP
ECK4055	soxR	b4063	4,277,469	4,277,933	+	SoxR	MerR
b2087	gatR	b4498	2,169,693	2,171,727	-	GatR	NA

- Other genomic features can be manipulated
- Example: transcription units reported by Conway et al., 2014

```
TUs_Conway <- read.table("Conway_2014.tsv")          # load data

TUs_updated <- TUs_Conway %>%                        # extract TU coordinates and strand
  [some parsing] %>%
  mutate(start = convert_coords(left)                # update coordinates to current
         stop = convert_coords(right)                # genome version
         genes = what_genes(start, stop, strand))    # get genes contained in TU
```

Operon	Genes
P-832279:T-835449	dinG-ybiB
P-835542:T-836826	ybiC
T-1304820:P-1306695	cls-yciU
P-1306788:T-1307015	yciY
P-1504781:T-1506984	ydcP
T-1506717:P-1507161	yncJ

start	stop	strand	genes
833,056	836,226	+	dinG, ybiB
836,319	837,603	+	hcxB, ybiE
1,306,796	1,308,671	-	yciU, clsA
1,308,764	1,308,991	+	yciY
1,506,757	1,508,960	+	rlhA
1,508,693	1,509,137	-	yncJ

Coming next...

- A dictionary of *E. coli*'s orthologs in *Salmonella enterica*
- Conversion of genes, TFs and regulatory network b/w both species

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References

- Conway T, Creecy JP, Maddox SM, Grissom JE, Conkle TL, et al (2014). Unprecedented high-resolution view of bacterial operon architecture revealed by RNA sequencing. mBio 2014;5:e01442-14.
- Wickham H, Girlich M (2022). tidyr: Tidy Messy Data. https://tidyr.tidyverse.org, https://github.com/tidyverse/tidyr